

① T is Hermitian operator on a Hermitian space V . Let $v \in V$.
Suppose $T^n v = 0$ for some $n > 1$. Show $Tv = 0$. (What happens when $n=2$?)

② Let v, w be eigenvectors of a normal operator T on a Hermitian space, with distinct eigenvalues λ, μ respectively.
Show $\langle v, w \rangle = 0$ (Spectral thm NOT allowed!)

③ Let A, B be unitary matrices with $AB = BA$. Show that there is a matrix P such that $P^* A P$ and $P^* B P$ are both diagonal matrices.

④ Let T be a normal operator on a Hermitian space

If each eigenvalue of T _____ then T is Hermitian.
If each eigenvalue of T _____ then T is unitary.

} Fill in the blanks and explain.

⑤ Let T be a linear operator on an inner product space V

Let W be a T -invariant subspace.

(a) See quickly that $W^\perp$ is T^* -invariant.

(b) If T is normal, show that $W^\perp$ is T -invariant.

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Possible hint.

{ Extend an orthonormal basis $e_1, \dots, e_k$ of W to get an _____ $e_1, \dots, e_k, f_1, \dots, f_l$ of V .
Consider $\sum_{i=1}^k \langle Te_i, Te_i \rangle$.